

Decentralized Autonomous Organizations (DAOs)

The Transition from Paper Legal Entities to On-Chain Institutional Architecture.



The Institutional Shift

1.

Joint-Stock Corporations

Legal
Enforcement

Managed by paper charters, boards of directors, and court jurisdictions.

Geographic
Access

Rigid borders (e.g., Delaware corps) restricting global talent and capital.

Execution
Speed

Slow, expensive bureaucratic overhead reliant on human executives.

2.

Internet-Native DAOs

Legal
Enforcement

Governed by immutable smart contracts executing on a public blockchain.

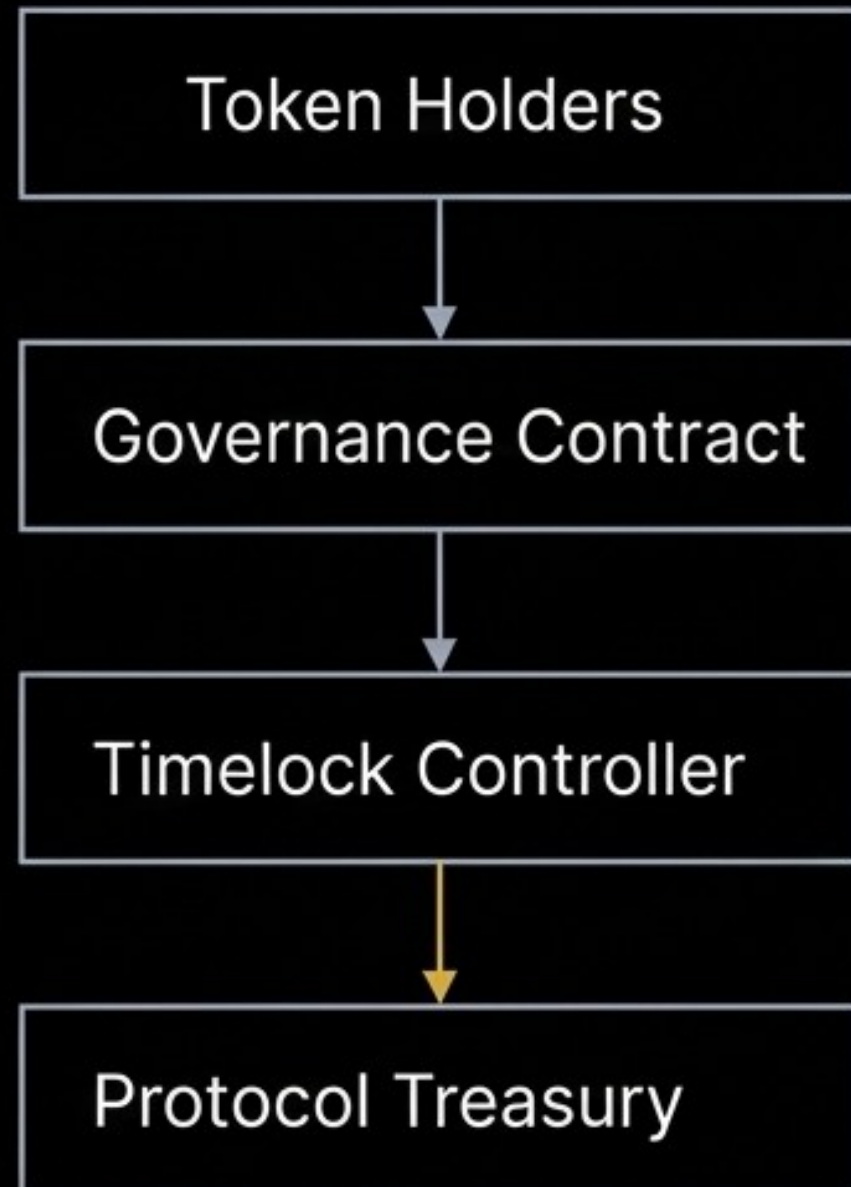
Geographic
Access

Global, permissionless access via cryptographic wallet signatures.

Execution
Speed

Autonomous bytecode execution with zero reliance on physical middlemen or CEOs.

Core DAO Architecture



Token Holders

Distributed community signaling consensus via wallet signatures.

Governor Engine

The core ledger that records proposals, counts votes, and verifies quorum.

Timelock Controller

The mandatory time-delay buffer for pending state changes.

Key Insight: Zero human intervention. Once a vote passes, the virtual machine executes the payload autonomously.

The Proposal Lifecycle: Off-Chain to On-Chain

Phase 1: Gasless Off-Chain

Step 1: Request for Comments (RFC) on open Discourse forums.

Step 2: Snapshot Temperature Check. Token holders sign gasless cryptographic messages to prove ownership and gauge initial consensus.

Phase 2: Gas-Intensive On-Chain

Step 3: On-chain propose() via GovernorBravo (requires high token threshold to prevent spam).

GovernorBravo propose() Payload

targets: Smart contract addresses to call.

values: Amount of native ETH to send.

signatures: Interface format (e.g., transfer(address,uint256)).

calldatas: Hexadecimal ABI-encoded parameters.

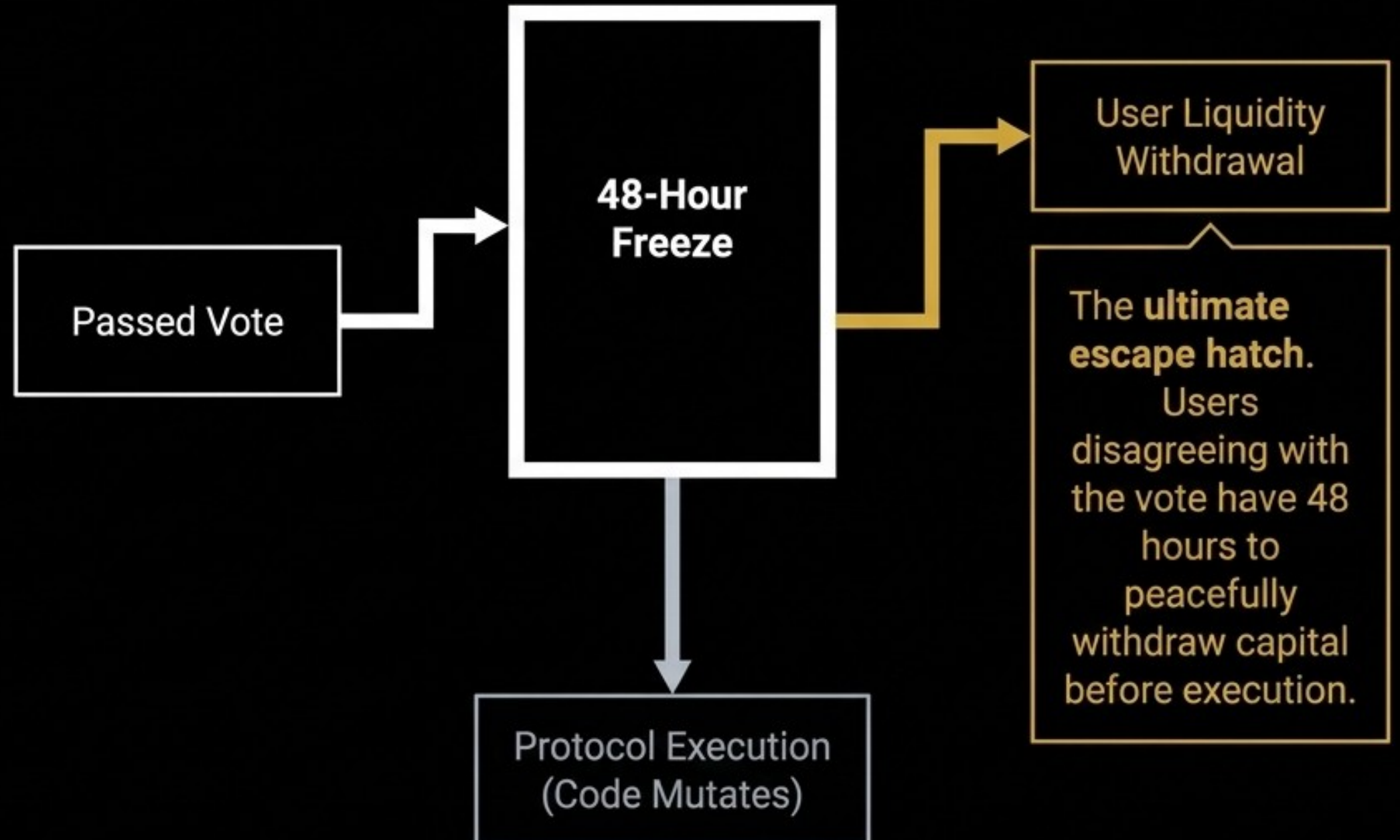
Consensus Mechanics: Quorum & The Timelock

The Snapshot Block

Voting power is permanently locked to a historical block height before the vote begins. This mathematically disarms immediate flash-loan borrowing attacks.

Mathematical Passage

Requires both Majority Approval (For > Against) AND a Quorum Threshold (typically 4%-10% of circulating supply).



The Voting Dilemma: Linear vs. Quadratic

Linear Plutocracy

$$V = T$$

1 Token = 1 Vote

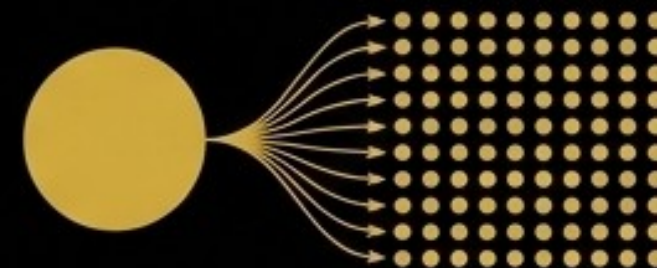
Power scales perfectly with capital. Results in The **Whale Dilemma**, where a single VC outvotes tens of thousands of grassroots users, leading to extreme voter apathy.

Quadratic Voting (QV)

$$V = \sqrt{T}$$

Voice = Square Root of Tokens

Empowers the grassroots by penalizing concentrated wealth. (e.g., 100 users with 100 tokens = 1,000 votes. 1 whale with 10,000 tokens = 100 votes).

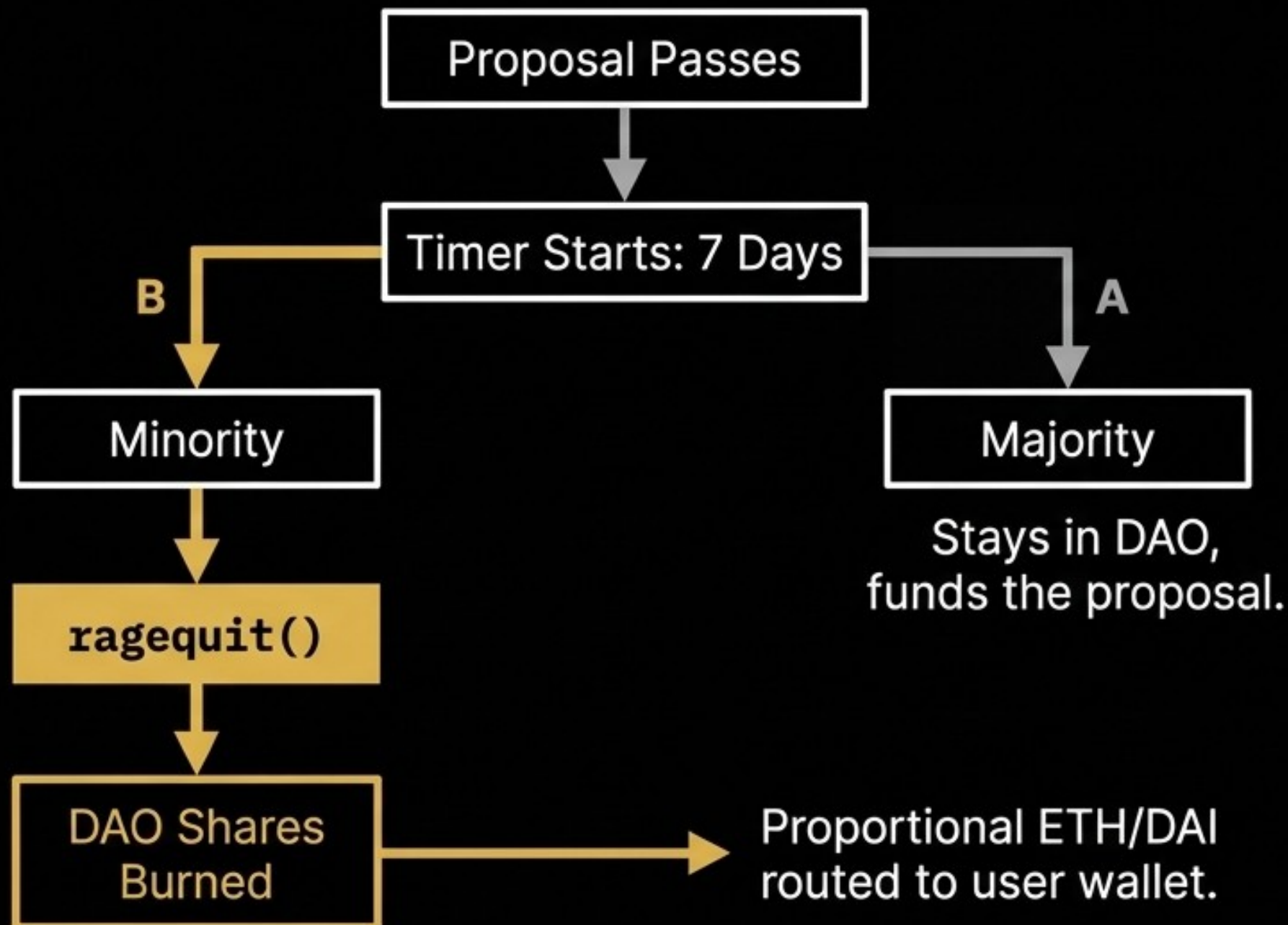


Whale

Fake Wallets

Sybil Attack Vulnerability:
On an anonymous blockchain, whales simply split their tokens across fake wallets to bypass the square-root penalty.

Minority Protection: The MolochDAO Architecture



The Philosophy

Ameen Soleimani's design to defeat coordination failure (Moloch) and majority tyranny.

The Mechanism

Members rejecting a proposal can call **ragequit()** before execution. Their shares are burned, and they extract their exact proportional slice of the treasury reserves peacefully.

Governance Threat Matrix: Attack Vectors

Case 1: Beanstalk Farms (2022)

Vector: Flash Loan Takeover (\$182M Drained)

- Borrow \$1B flash loan
- Buy 67% supermajority
- Pass malicious BIP-18
- Drain treasury & repay loan in single block

Fatal Failure Point: Bypassing the mandatory Timelock/Snapshot block.

Case 2: Build Finance (2022)

Vector: Low Quorum Hijacking (Infinite Minting)

- Capitalize on community apathy
- Buy minimal tokens to meet 5% quorum
- Submit disguised proposal
- Extract hidden minting keys with zero opposition

Fatal Failure Point: Lack of dynamic quorum thresholds or a Security Council veto.

The Scalability Wall (Transitioning to Layer 2)

